

Single Core Copper Conductor, XLPE insulated

Lead Sheathed and HDPE Sheath
(CU/XLPE/LEAD/HDPE)

Voltage Grade

76 / 132 (145) KV

Description

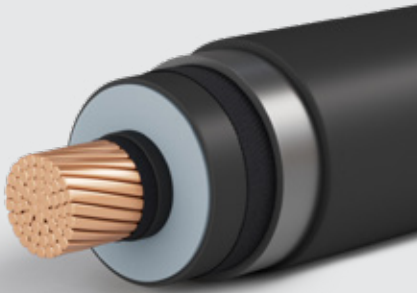
Stranded circular or segmental compacted copper conductor Filled with swelling powder between conductor layers, Semi-conductive water blocking tape over conductor to protect the Conductor from longitudinal water penetration, semi-conducting layer as conductor screen, XLPE insulated, semi-conducting layer as non metallic insulation screen, Semi-conductive water blocking tape to protect the screen area from longitudinal water penetration, extruded lead sheathed with a suitable thickness to withstand the required earth fault current, and also act as a radial water pentration protection for the cable and HDPE sheathed with graphite coating or extruded semi-conducting layer for the D.C Sheath test.

Application

These cables are generally suitable for direct burial or for installation on trays or in ducts.

Technical Data

Relevant Standard	IEC 60228, 60840, IEC 60229 and 60811
Conductor	Plain copper conductor class 2 Circular Compacted (applied for C.S.A up to 800 mm2) or Segmental Compacted (Applied from C.S.A = 1000 mm2 and above) Five segments of compacted conductor in sector shape of 72° are assembled together with separation of non-metallic tapes to reduce the skin effect which reduce the Rac.
Conductor Screen	Extruded Semiconductive Compound
Insulation	Extruded XLPE Compound Rated 90° C
Insulation Screen	Non Metallic part: Extruded Semiconductive Compound (Bonded) to the insulation Metallic part: Extruded Lead Sheath with a suitable thickness Conductor screen, XLPE insulation and non metallic insulation screen are applied at the same time using triple head extruder.
Outer Sheath	An extruded layer of HDPE shall be applied over the metallic layers for protection against chemical reaction & mechanical protection. Extruded semi conductive layer or graphite coating shall be applied over the outer sheath for D.C testing purpose
Water Pentration layers	i) Longitudinal Water Penetration: 1) the conductor: the conductor layers filled with swelling powder and overlaped semiconductive water blocking tape over the conductor 2) The Metallic part of the insulation screen: An overlaped semiconductive water blocking tape before the metallic screen ii) Radial Water Penetration: Extruded lead sheath applied before the PE outer Sheath
Outer Sheath Color	Black (default color), or any other color according to customer request



HIGH VOLTAGE CABLES

Maximum Operating Temp. 90° C

Minimum bending Radius (16-18) x outer diameter

Packing Condition Our standard length on Steel Drum / or according to customer request

Product Code	Nominal Cross Sectional Area	Conductor			Operating Capacitance	Inductance		Thickness of Conductor Screen	Thickness of Insulation	Thickness of Conductor Screen	Thickness of Insulation	Approx. Overall Diameter	Approx. Weight		
		DC at 20 °C	AC at 90 °C			Flat seperated	Trefoil								
			Flat seperated	Trefoil											
															
															
mm²	Ω/km	Ω/km	Ω/km	μF/km	mH/km	mH/km	mm	mm	mm	mm	mm	kg/km			
Electrical & Constrctional data (Nominal Values)															
PH-CG20-XD-01-AH	240 R	0.0754	0.0970	0.0972	0.1420	0.6603	0.4754	1.2	18.0	1.2	4.5	75.7	11940		
PH-CG21-XD-01-AH	300 R	0.0601	0.0777	0.0780	0.1512	0.6428	0.4579	1.2	18.0	1.2	4.5	77.7	12630		
PH-CG22-XD-01-AH	400 R	0.0470	0.0613	0.0617	0.1623	0.6230	0.4381	1.2	18.0	1.2	4.5	80.2	13605		
PH-CG23-XD-01-AH	500 R	0.0366	0.0485	0.0491	0.1757	0.6039	0.4191	1.2	18.0	1.2	4.5	83.3	14890		
PH-CG24-XD-01-AH	630 R	0.0283	0.0384	0.0392	0.1903	0.5865	0.4016	1.2	18.0	1.2	4.5	86.7	16410		
PH-CG25-XD-01-AH	800 R	0.0221	0.0311	0.0322	0.2080	0.5700	0.3852	1.2	18.0	1.2	4.5	90.9	18530		
PH-CM26-XD-01-AH	1000 S	0.0176	0.0246	0.0249	0.2403	0.5512	0.3663	1.5	18.0	1.2	4.5	99.0	22120		
PH-CM27-XD-01-AH	1200 S	0.0151	0.0218	0.0222	0.2506	0.5441	0.3592	1.5	18.0	1.2	4.5	101.6	23805		

Product Code	Nominal Cross Sectional Area	Laying conditions: Trefoil formation			Laying conditions: Flat formation		
		Type of Earthing Bonding System	Direct burial	in Air (Shaded)	Type of Earthing Bonding System	Direct burial	in Air (Shaded)
							
			$p_t=120$ $T=35\text{ }^{\circ}\text{C}$	$T=40\text{ }^{\circ}\text{C}$		$p_t=120$ $T=35\text{ }^{\circ}\text{C}$	$T=40\text{ }^{\circ}\text{C}$
Continuous current ratings (load factor = 100%) for one circuit in operation (Amperes)							
PH-CG20-XD-01-AH	240 R	Double End Bonded	436	593	Single End Bonded or Cross Bonding	470	665
PH-CG21-XD-01-AH	300 R		489	675		531	762
PH-CG22-XD-01-AH	400 R		552	775		606	882
PH-CG23-XD-01-AH	500 R		622	891		691	1024
PH-CG24-XD-01-AH	630 R		697	1018		786	1186
PH-CG25-XD-01-AH	800 R		771	1153		886	1364
PH-CM26-XD-01-AH	1000 S		853	1324		1015	1599
PH-CM27-XD-01-AH	1200 S		898	1414		1088	1729

The above data is approximate and subject to manufacturing tolerances

Note:-

Flame retardant PVC sheathed or MDPE, LLDPE Sheathed are available according to customer request
The Metallic Screen is designed to carry an earth fault cureent =40 KA per one seond per three pahses, any other design is also available as per customer request